Reg. No. | | | | | | | | | | |

[14CE82]

EIGHTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

ADVANCED CONCRETE TECHNOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions to the candidates:

- Answer any **FIVE** questions selecting at least **TWO** from each part.
- Assume the missing data if any and indicate clearly.
- Use of IS 10262 is allowed

PART - A

1. a. Explain the meaning of high performance concrete and how it is different from High strength concrete. Discuss a typical mix as an example. **08**

   b. Explain at least four properties of high performance concrete. **06**

2. a. What are accelerators and why they are required? Name at least two such compounds **06**

   b. What are super plasticizers and explain their working mechanism with reference to sketches. Name any two such compounds. **08**

3. a. What is the meaning of RMC and why it is preferred in the present days compared to conventional concrete? **06**

   b. Explain the problems normally encountered in RMC use and how they are overcome during various stages. **08**

4. a. Explain the various provisions in the BIS: 456-2000 with regards to concrete sampling, testing and strength requirements. **08**

   b. In concrete mix design, explain how the target strength is determined and the various provisions for standard deviation values and test results for acceptable values **06**

PART - B

5. a. Explain the necessity of testing hardened concrete. Explain the effect of end conditions of specimen, capping and H/D ratio on strength of concrete. **08**

   b. Explain with relevant sketches how Saint Venant's principle is useful in accessing the strength of concrete. **06**

6. a. What is the meaning of high density concrete and how it is achieved? What are its applications? **08**

   b. Discuss two properties of High Density concrete and name two short comings of it. **06**

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7. a. Explain the meaning of Non-destructive testing of concrete and why it is preferred compared to conventional method. Explain the Principe of the test. **08**

b. Explain why the Nuclear method of concrete testing is preferred for concrete and what its limitations. **06**

8. a. Explain the reasons how the durability of concrete is reduced due to permeability, chemical attack and acid attack. What are the counter measures to minimize it. **10**

b. What is efflorescence and what are its effects. **04**